

## SAFETY DATA SHEET

Revision Date 29-March-2024

Revision Number 3

### 1. Identification

**Product Name** 4-Acetamidobenzenesulfonyl chloride

**Cat No. :** L00866

**CAS-No** 121-60-8

**Synonyms** 4-Acetamidobenzenesulfonyl chloride; ASC

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

**Skin Corrosion/Irritation**

Category 1 B

**Serious Eye Damage/Eye Irritation**

Category 1

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Causes severe skin burns and eye damage

**Precautionary Statements****Prevention**

Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF SWALLOWED: Rinse mouth. Do NOT induce vomiting  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Wash contaminated clothing before reuse

**Storage**

Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component                                  | CAS-No   | Weight % |
|--------------------------------------------|----------|----------|
| Benzenesulfonyl chloride, 4-(acetylamino)- | 121-60-8 | >95      |

### 4. First-aid measures

|                                        |                                                                                                                                                                                                                                                                                                 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                               |
| <b>Eye Contact</b>                     | Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Keep eye wide open while rinsing.                                                                                                                             |
| <b>Skin Contact</b>                    | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.                                                                                                                                                             |
| <b>Inhalation</b>                      | Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.                                                                               |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Drink plenty of water. Immediate medical attention is required.                                                                                                                                                  |
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                           |

### 5. Fire-fighting measures

|                                         |                                                            |
|-----------------------------------------|------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Dry sand. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                   |
| <b>Flash Point</b>                      | > 112 °C / > 233.6 °F                                      |
| <b>Method -</b>                         | No information available                                   |
| <b>Autoignition Temperature</b>         | Not applicable                                             |
| <b>Explosion Limits</b>                 |                                                            |
| <b>Upper</b>                            | No data available                                          |
| <b>Lower</b>                            | No data available                                          |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                   |
| <b>Sensitivity to Static Discharge</b>  | No information available                                   |

**Specific Hazards Arising from the Chemical**

The product causes burns of eyes, skin and mucous membranes.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides. Thermal decomposition can lead to release of irritating gases and vapors. Hydrogen chloride gas.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**NFPA**

**Health**  
3

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                  |                                                                                                                                                              |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Evacuate personnel to safe areas. Use personal protective equipment as required. Avoid contact with skin, eyes or clothing.                                  |
| <b>Environmental Precautions</b> | Do not allow material to contaminate ground water system. Should not be released into the environment. See Section 12 for additional Ecological Information. |

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Do not breathe dust. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not ingest. If swallowed then seek immediate medical assistance. |
| <b>Storage.</b> | Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place. Store under an inert atmosphere. Incompatible Materials. Strong oxidizing agents. Strong bases.                                      |

## 8. Exposure controls / personal protection

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b>  | This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.                                                                                                                                                                                                                                                                                                             |
| <b>Engineering Measures</b> | Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.<br>Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source |

**Personal protective equipment**

**Eye Protection**  
**Hand Protection**

Goggles  
 Protective gloves

| Glove material                                      | Breakthrough time                 | Glove thickness | Glove comments         |
|-----------------------------------------------------|-----------------------------------|-----------------|------------------------|
| Natural rubber<br>Nitrile rubber<br>Neoprene<br>PVC | See manufacturers recommendations | -               | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

Wear a NIOSH/MSHA or European Standard EN 149 approved full-facepiece airline respirator in the positive pressure mode with emergency escape provisions.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                               |                               |
|-----------------------------------------------|-------------------------------|
| <b>Physical State</b>                         | Powder Solid                  |
| <b>Appearance</b>                             | Off-white                     |
| <b>Odor</b>                                   | acetic acid                   |
| <b>Odor Threshold</b>                         | No information available      |
| <b>pH</b>                                     | No information available      |
| <b>Melting Point/Range</b>                    | 145 - 148 °C / 293 - 298.4 °F |
| <b>Boiling Point/Range</b>                    | No information available      |
| <b>Flash Point</b>                            | > 112 °C / > 233.6 °F         |
| <b>Evaporation Rate</b>                       | Not applicable                |
| <b>Flammability (solid,gas)</b>               | No information available      |
| <b>Flammability or explosive limits</b>       |                               |
| Upper                                         | No data available             |
| Lower                                         | No data available             |
| <b>Vapor Pressure</b>                         | No information available      |
| <b>Vapor Density</b>                          | Not applicable                |
| <b>Specific Gravity</b>                       | No information available      |
| <b>Solubility</b>                             | No information available      |
| <b>Partition coefficient; n-octanol/water</b> | No data available             |
| <b>Autoignition Temperature</b>               | Not applicable                |
| <b>Decomposition Temperature</b>              | No information available      |
| <b>Viscosity</b>                              | Not applicable                |
| <b>Molecular Formula</b>                      | C8 H8 Cl N O3 S               |
| <b>Molecular Weight</b>                       | 233.67                        |

## 10. Stability and reactivity

**Reactive Hazard**

None known, based on information available

|                                         |                                                                                                                                                                         |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Stability</b>                        | Moisture sensitive.                                                                                                                                                     |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Exposure to moist air or water.                                                                                                     |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Strong bases                                                                                                                                   |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Sulfur oxides, Thermal decomposition can lead to release of irritating gases and vapors, Hydrogen chloride gas |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                                                                |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                                                                           |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component                                  | LD50 Oral           | LD50 Dermal               | LC50 Inhalation |
|--------------------------------------------|---------------------|---------------------------|-----------------|
| Benzenesulfonyl chloride, 4-(acetylamino)- | >3200 mg/kg ( Rat ) | LD50 = 6750 mg/kg ( Rat ) | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                                  | CAS-No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------------------------------------|----------|------------|------------|------------|------------|------------|
| Benzenesulfonyl chloride, 4-(acetylamino)- | 121-60-8 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

| Component                                  | Freshwater Algae | Freshwater Fish                          | Microtox   | Water Flea |
|--------------------------------------------|------------------|------------------------------------------|------------|------------|
| Benzenesulfonyl chloride, 4-(acetylamino)- | Not listed       | LC50: > 2 mg/L, 96h static (Danio rerio) | Not listed | Not listed |

**Persistence and Degradability** Persistence is unlikely

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility.

| Component                                  | log Pow |
|--------------------------------------------|---------|
| Benzenesulfonyl chloride, 4-(acetylamino)- | 2.05    |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN3261  
 Proper Shipping Name consumer commodity Corrosive solid, acidic, organic, n.o.s.  
 Technical Name 4-(Acetylamino) benzenesulfonyl chloride  
 Hazard Class 8  
 Packing Group II

#### TDG

UN-No UN3261  
 Proper Shipping Name Corrosive solid, acidic, organic, n.o.s.  
 Hazard Class 8  
 Packing Group II

#### IATA

UN-No UN3261  
 Proper Shipping Name Corrosive solid, acidic, organic, n.o.s.  
 Hazard Class 8  
 Packing Group II

#### IMDG/IMO

UN-No UN3261  
 Proper Shipping Name Corrosive solid, acidic, organic, n.o.s.  
 Hazard Class 8  
 Packing Group II

### 15. Regulatory information

#### International Inventories

| Component                                  | CAS-No   | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|--------------------------------------------|----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Benzenesulfonyl chloride, 4-(acetylamino)- | 121-60-8 | X   | -    | X    | ACTIVE                                        | 204-485-1 | -      | -   |

| Component                                  | CAS-No   | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|--------------------------------------------|----------|-------|----------|------|------|------|------|-------|-------|
| Benzenesulfonyl chloride, 4-(acetylamino)- | 121-60-8 | X     | KE-00083 | X    | X    | X    | -    | X     | -     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**ENCS** - Japanese Existing and New Chemical Substances  
**AICS** - Australian Inventory of Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

#### Other International Regulations

Authorisation/Restrictions according to EU REACH Not applicable

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                                  | CAS-No   | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------------------------------------|----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Benzenesulfonyl chloride, 4-(acetylamino)- | 121-60-8 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                                  | CAS-No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------------------------------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Benzenesulfonyl chloride, 4-(acetylamino)- | 121-60-8 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

### 16. Other information

**Prepared By** Product Safety Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Revision Date** 29-March-2024  
**Print Date** 29-March-2024  
**Revision Summary** New emergency telephone response service provider.

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**